Redone derivation on page 189 + footnote 15:

The Bellman equation implies a. $\tilde{p} = c + \int_{\underline{p}}^{B} v(p')dH(p')$. Applying the optimal decision rule for p, namely that if $p \geq \tilde{p}$ $v(p') = \tilde{p}$ and if $p \leq \tilde{p}$ $v(p) = p$ we get:

1. $\int_{\underline{p}}^{B} v(p')dH(p') = \int_{\underline{p}}^{\tilde{p}} p'dH(p') + \int_{\tilde{p}}^{B} \tilde{p}dH(p')$

We note:

2. $\tilde{p} = \int_{\underline{p}}^{\tilde{p}} \tilde{p}dH(p') + \int_{\tilde{p}}^{B} \tilde{p}dH(p')$

Thus, combining 1 and 2 with a. we get: $\int_{\underline{p}}^{\tilde{p}} (\tilde{p} - p')dH(p') = c$

Applying integration by parts $\int udv = uv - \int vdu$, with $dv = dH(p')$ and $u = \tilde{p} - p'$:

$(\tilde{p} - p')H(p')|_{\underline{p}}^{\tilde{p}} + \int_{\underline{p}}^{\tilde{p}} H(p')dp' = c$

which upon evaluating implies:

3. $\int_{\underline{p}}^{\tilde{p}} H(p')dp' = c$

Let

$E[p|p \leq \tilde{p}] = \int_{\underline{p}}^{\tilde{p}} (1 - H(p'))dp'$

which can be rewritten as :

4. $E[p|p \leq \tilde{p}] = \tilde{p} - \underline{p} - \int_{\underline{p}}^{\tilde{p}} H(p')dp'$

Combining 3 and 4 we get:

$c + E[p|p \leq \tilde{p}] = \tilde{p} - \underline{p}$

or:

$c + \underline{p} + E[p|p \leq \tilde{p}] = \tilde{p}$

This states that the difference between the reservation price and the lowest possible price equals the cost of one additional round of search plus the mean of prices we will accept, that is prices lower than the reservation price.